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Two-dimensional ferromagnetic semiconductors of rare-earth Janus 2H-GdIBr monolayers with large valley polarization†

Cunquan Li and Yukai An *

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Based on a rare-earth Gd atom with 4f electrons, through first-principles calculations, we demonstrate that a Janus 2H-GdIBr monolayer exhibits an intrinsic ferromagnetic (FM) semiconductor character with an indirect band gap of 0.75 eV, a high Curie temperature T_c of 260 K, a significant magnetic moment of $8\mu_B$ per f.u. (f.u. = formula unit), in-plane magnetic anisotropy (IMA) and a large spontaneous valley polarization of 118 meV. The MAE, inter-atomic distance or angle, and T_c can be efficiently modulated by in-plane strains and charge carrier doping. Under a strain range from −5% to 5% and charge carrier doping from −0.3 e to 0.3 e per f.u., the system still retains its FM ordering and the corresponding T_c can be modulated by strains from 233 K to 281 K and by charge carrier doping from 140 K to 245 K. Interestingly, under various strains, the matrix element differences (d_{z^2}, d_{yz}), ($d_{x^2-y^2}$, d_{xy}) and (p_x, p_y) of Gd atoms dominate the MAE behaviors, which originates from the competition between the contributions of the Gd-d orbitals, Gd-p orbitals, and p orbitals of halogen atoms based on the second-order perturbation theory. Inequivalent Dirac valleys are not energetically degenerate due to the time-reversal symmetry breaking in the Janus 2H-GdIBr monolayer. A considerable valley gap between the Berry curvature at the K and K′ points provides an opportunity to selectively control the valley freedom and states. External tensile (compressive) strain further increases (decreases) the valley gap up to a maximum (minimum) value of 158 (37) meV, indicating that the valley polarization in the Janus 2H-GdIBr monolayer is robust to external strains. This study provides a novel paradigm and platform to design spintronic devices for next-generation quantum information technology.

1. Introduction

As described by the Mermin–Wagner theorem,[1] the long-range magnetic order of two-dimensional (2D) systems at finite temperature is forbidden by strong fluctuations. Recent studies suggest that magnetic anisotropy opens a gap in the spin-wave spectrum and suppresses the effect of thermal fluctuations. Consequently, when a 2D system exhibits intrinsic anisotropy caused by spin–orbit coupling (SOC), its long-range magnetic order can be well preserved, which has been observed in the CrI_3 monolayer and $Cr_2Ge_2Te_6$ bilayer with a Curie temperature (T_c) of 45 K (ref. 2) and 28 K,[3] respectively. On the other hand, it has been proved that T_c can be moderately increased by various methods, including carrier doping,[4] strain,[5] electric field,[6] and so on. However, these extrinsic methods and low T_c are not ideal for the advancement of 2D magnetic devices in spintronics. Therefore, the exploration of intrinsic 2D ferromagnetic semiconductors with high T_c is an urgent need.[7] Recently, various 2D materials with hexagonal lattice structures, especially for group-VI transition-metal dichalcogenides (TMDs), have opened the possibility of practical implementation of valleytronics. In monolayer TMDs, the conduction-band minima (CBM) and valence-band maxima (VBM) are located at two degenerate but inequivalent K and K′ valleys of a hexagonal Brillouin zone due to spatial inversion symmetry breaking.[8] Therefore, the so-called K-valley materials have attracted much attention due to their potential applications in nanoscale valleytronic devices.[9] Most 2D valley materials do not exhibit spontaneous valley polarization due to their conservation of inversion symmetry. In general, breaking the inversion symmetry also lifts the spin degeneracy of an energy band due to the presence of SOC effects. The introduction of valley polarization is necessary to distinguish and manipulate the carriers in the specified valleys. In other

Key Laboratory of Display Materials and Photoelectric Devices, Ministry of Education, Tianjin Key Laboratory for Photoelectric Materials and Devices, National Demonstration Center for Experimental Function Materials Education, School of Material Science and Engineering, Tianjin University of Technology, Tianjin, 300384, China. E-mail: ykan@tjut.edu.cn

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respects, charge carriers can also be distinguished by their spin moments. Thus, many peculiar properties will also arise, such as the quantum valley anomaly Hall effect.[10] If the intrinsic ferromagnetic ordering can introduce spontaneous valley polarization, the so-called ferro-valley materials are proposed. Until now, only a few ferro-valley materials have been predicted, including $TiVI_6$,[11] VSe_2 [12] and $VAgP_2Se_6$ [13] monolayers, *etc*. However, these systems are more focused on the d-electron, and studies on f-electron systems are relatively fewer. Inspired by the significant magnetic moments and high magneto-crystal anisotropy typically associated with rare-earth (RE) elements,[14] introducing RE elements is expected to largely increase the T_c and magnetization of these systems. Also, experiments on ultrathin 2D RE nanomaterials have been reported and they exhibited great potential in various applications for electronic and magnetic devices.[15]

In particular, as a prototype RE metal with a half-filled 4f shell, the gadolinium element has been chosen for its special room temperature (RT) ferromagnetism among the RE elements. Early in 1965, Mee and Corbett synthesized layered ferromagnetic GdI_2 bulk crystals with a T_c above RT.[16,17] However, the corresponding studies on Gd-based monolayers have only been reported recently. Prominently, Wang *et al.* demonstrated theoretically that the GdI_2 monolayer may be easily exfoliated from its layered bulk and maintain an intrinsic T_c above RT.[18] On the other hand, Zhang *et al.* proved that the $GdBr_2$ monolayer shows excellent thermal/dynamic stabilities with the strain-driven tunable easy axis from in-plane to out-of-plane.[19] Motivated by the quite exciting quantum phenomena arising from a broken mirror symmetry in 2D Janus monolayer structures and following the research studies on GdX_2 (X = I, Br) monolayers,[20] the Janus 2H-GdIBr monolayer can be well built by replacing the atomic layer on one side. In fact, the Janus 2H-GdIBr monolayer has not yet been prepared experimentally. However, many research studies on the direct synthesis or replacement of atomic layers to obtain the Janus system have been reported. Very recently, the Janus MSSe (M = W, Mo) monolayer has been synthesized by the room-temperature replacement method based on the MX_2 (X = S, Se) monolayer.[21,22] Gan *et al.* reported a one-pot chemical vapor deposition synthesis of the large-area Janus MoSSe monolayer.[23] Based on our discussion above, a possible preparation route for experimentally obtaining the Janus 2H-GdIBr monolayer is as follows: first, the GdI_2 monolayer is obtained by exfoliation from its layered bulk crystal. Next, the top-layer I atoms are stripped off and replaced with H atoms using a remote hydrogen plasma (GdIH), followed by the replacement of the H atoms by Br atoms using subsequent thermal bromination (forming the Janus 2H-GdIBr monolayer) by controlling the kinetics and thermodynamics of the reaction.

In this work, the stability and electronic, magnetic, and valley properties of the RE-based Janus 2H-GdIBr monolayer are investigated systematically. Notably, the Janus 2H-GdIBr monolayer can be easily exfoliated from the parent layered bulk crystal and it exhibits good thermodynamic and kinetic stability as well as a high T_c of 260 K, a large magnetic moment of $8\mu_B$ per f.u., a spontaneous valley polarization of 118 meV and in-plane magnetic anisotropy (−420 μeV per f.u). The Berry curvature of K and K′-valleys shows opposite signs, and the non-zero Berry phase leads to a non-zero anomalous Hall conductivity. These findings strongly suggest the potential application of the Janus 2H-GdIBr monolayer in valley electronic devices.

2. Computational methods

All calculations are performed based on generalized density-functional theory (DFT) using the Perdew–Burke–Ernzerhof (PBE) function in the generalized gradient approximation (GGA),[24,25] implemented in the Vienna *ab initio* simulation package (VASP).[26–28] A vacuum space of 18 Å is set to avoid the interaction between monolayers, and an energy cut-off of 500 eV is set for the plane-wave basis set. The Brillouin zone is sampled using a converged Γ-centered 8 × 8 × 1 *k*-mesh for structural relaxation and 18 × 18 × 1 for the electronic calculations. The standard pseudopotential is used, and the valence electron configuration is considered as $5s^25p^64f^75d^16s^2$ for Gd,[29] $4s^24p^5$ for Br, and $5s^25p^5$ for I. The crystal structure of the Janus 2H-GdIBr monolayer is fully relaxed, and the convergence criteria of energy and forces are 10^{-7} eV and 0.01 eV $Å^{-1}$, respectively. The rotationally invariant local spin-density approximation (LSDA) + U method is used to handle the strongly correlated corrections to the Gd 4f electrons, and the on-site Coulomb interaction parameter (U) and exchange interaction parameter (J) are set at 9.20 and 1.20 eV, respectively.[30,31] The phonon dispersion spectrum of the Janus 2H-GdIBr monolayer is obtained using the PHONOPY code[32,33] based on the density-functional perturbation theory (DFPT)[34] using a 4 × 4 × 1 supercell. The VASPKIT code is used to process the VASP data.[35] An *ab initio* molecular dynamic (AIMD) simulation is adopted in the NVT ensemble based on the Nosé-Hoover thermostat[36] which controlled the temperature of the system at 300 K with a total of 10 ps at 2 fs per step. The Monte Carlo simulation package MCSOLVER[37] is used to estimate the T_c based on the classical Heisenberg model. Starting with the ferromagnetic configuration, 80 000 sweeps are used to fully thermalize the system to equilibrium, and all statistics are obtained from the 720 000 scans that followed. The Berry curvature and anomalous Hall conductivity (AHC) of the Janus 2H-GdIBr monolayer are calculated using maximally localized Wannier functions implemented in the WANNIER90[38] and WannierTools packages.[39] VASPBERRY[40] is used to study its optical properties.

3. Results and discussion

The top and side views of the crystal structures for the Janus 2H-GdIBr monolayer are shown in Fig. 1a. The Janus 2H-GdIBr monolayer consists of three atomic layers, where the Gd atomic layer is sandwiched between two different halogen

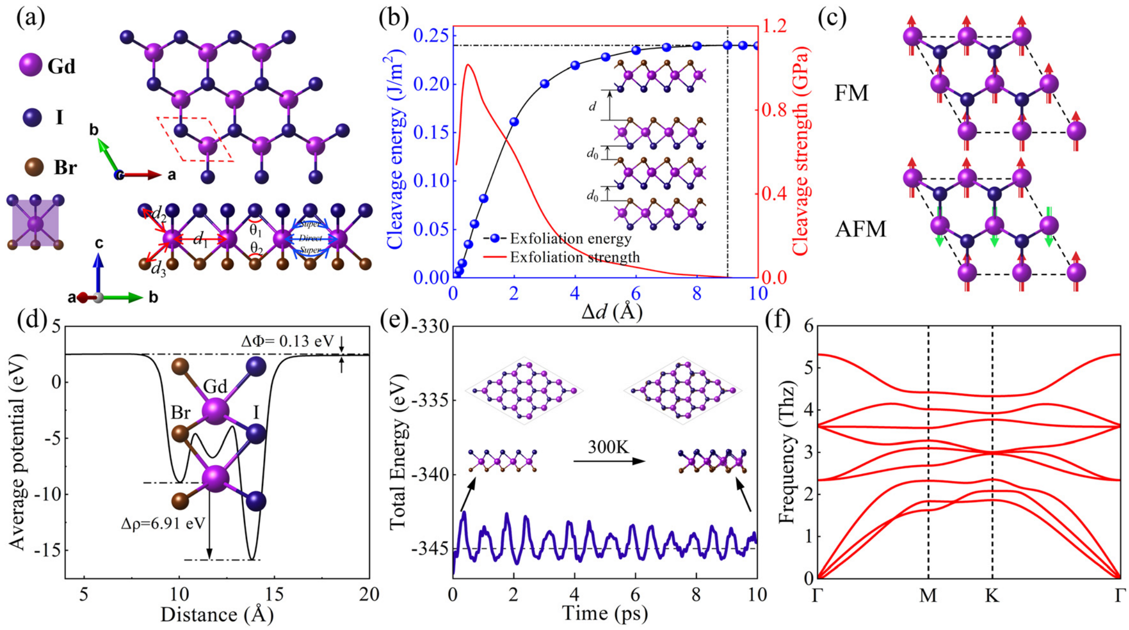

Fig. 1 (a) Top and side views of the Janus 2H-GdIBr monolayer with the nearest Gd–Gd distance d_1, Gd–I(Br) distance d_2 (d_3), and Gd–I(Br)–Gd bond angle $\theta_1(\theta_2)$. (b) The dependence of cleavage energy and cleavage strength on the separation distance ($\Delta d = d - d_0$) for the Janus 2H-GdIBr monolayer, where d_0 indicates the equilibrium vdW gap in the bulk crystal. The inset shows the calculated exfoliation process. (c) The FM and AFM orders for the Janus 2H-GdIBr monolayer. Red and green arrows denote spin-up and spin-down orientations, respectively. (d) Planar average electrostatic potential energy of the Janus 2H-GdIBr monolayer. (e) The actuation of total energy and snapshots of geometric structures for the Janus 2H-GdIBr monolayer at 10 ps from AIMD simulations. (f) The phonon dispersion of the Janus 2H-GdIBr monolayer.

atomic layers. Meanwhile, the central Gd atom is a trigonal shape coordinated to six halogen atoms. The calculated relaxation equilibrium lattice constants and Gd–I and Gd–Br bond lengths for the Janus 2H-GdIBr monolayer are 4.019 Å ($a = b$), 3.121 Å, and 2.955 Å, respectively. The cleavage energy is calculated from a slab model to confirm the possibility of exfoliation of the Janus 2H-GdIBr monolayer from its layered bulk crystal. As shown in Fig. 1b, the energy difference increases with the increase of the separation distance ($\Delta d = d - d_0$). Eventually, it converges to 0.24 J m^{-2} at the Δd = 9 Å point, which is taken as the exfoliation energy and confirmed by the calculated cleavage strength (the first derivative of cleavage energy). The exfoliation energy is less than the experimental value of graphite (0.36 J m^{-2}),[41] suggesting the feasibility of exfoliating the Janus 2H-GdIBr monolayer experimentally. Due to the contribution of half-filled 4f and 5d shells, the calculated magnetic moment of Gd atom is 7.4μ_B per f.u. Different magnetic configurations including ferromagnetic (FM) and antiferromagnetic (AFM) states are considered, as shown in Fig. 1c, which are the same as those reported previously.[19,42] To investigate the magnetic ground state and configurations of the Janus 2H-GdIBr monolayer, the energies of FM and AFM states of a 2 × 2 × 1 supercell are compared. The energy of the FM state is lower than that of the AFM state by 152 meV per Gd [Fig. S1 of the ESI†], indicating the existence of strong FM coupling. As shown in Fig. S2a of the ESI,† with the system always remaining in the FM state, the variation in the total energy of FM and AFM states can be considered a quadratic function of strains. Charge carrier doping can also effectively modulate the total energy of FM and AFM states [Fig. S2b in the ESI†], but the system remains in the FM state. The calculated average electrostatic potential ($\Delta\rho$) along the z-axis is asymmetric with a potential drop of 6.9 eV, as shown in Fig. 1d. In the sandwich structure, the potential difference between the two sides implies a change in the work function. Obviously, on the I side, the potential energy is smaller and the work function is larger than those on the Br side, which is related to the larger electronegativity of the I atom. Moreover, the electrostatic potential difference $\Delta\Phi$ of 0.13 eV can be considered as the redistribution of charges in the Janus 2H-GdIBr monolayer. Fig. 1e shows the actuation of total energy for the Janus 2H-GdIBr monolayer during the whole AIMD simulations, and the insets are the snapshots of the geometry at 0 ps and 10 ps, respectively. Clearly, the simulated total free energy fluctuates within a small range, and no significant deformation is observed at the end of the simulation, indicating good thermal stability of the system. In addition, to further investigate the stability of the Janus 2H-GdIBr monolayer, the phonon dispersion of the Janus 2H-GdIBr monolayer with a 4 × 4 × 1 supercell is shown in Fig. 1f. The frequencies of all phonon modes within the Brillouin zone are positive, indicating that the Janus 2H-GdIBr monolayer is dynamically stable.

Fig. 2a shows the band structure of the Janus 2H-GdIBr monolayer with spin polarization (without considering SOC). The spin splitting leads to an indirect band gap of 0.75 eV, with the VBM at the K(K′) point and the CBM at the M point.

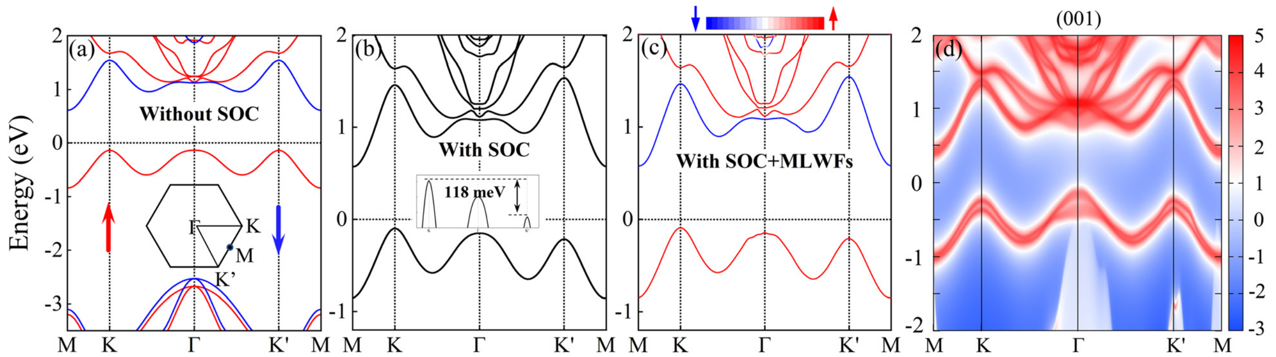

Fig. 2 The band structure of the Janus 2H-GdIBr monolayer (a) without SOC; (b) with SOC, and (c) with SOC + MLWFs, respectively. Red (blue) arrows represent spin up (down). (d) The local density of states of edge states for the Janus 2H-GdIBr monolayer projected on the (001) surface. The E_F is set to zero.

Around the Fermi level (E_F), the fully spin-polarized valence band and the opposite spin channel conduction band make the Janus 2H-GdIBr monolayer a bipolar magnetic semiconductor (BMS). The spin-resolved density of states shown in Fig. S3 of the ESI† demonstrates that the d orbitals of Gd atoms contribute to the VBM and CBM for the Janus 2H-GdIBr monolayer. The spin splitting around E_F does not appear for the I and Br atoms, implying that the magnetic moments of the Janus 2H-GdIBr monolayer are mainly from the Gd atoms. As shown in Fig. 2b, the SOC effect breaks the degeneracy between the K and K′ valleys in the valence band, namely, the energy of the K valley is higher than that of the K′ valley. This results in a large spontaneous valley polarization of 118 meV, which is equivalent to a huge external magnetic field of about 590–1180 T. The valley states at the K and K′ points are mainly contributed by the occupied $d_{x^2-y^2}$ and d_{xy} orbitals of the Gd atom and the Γ point is mainly distributed around the d_{z^2} orbital of the Gd atom, as shown in Fig. S4 of the ESI.† The spontaneous valley polarization can be attributed to the strong SOC effect combined with the magnetic exchange interaction of Gd-d electrons due to the breaking of time-reversal symmetry. To further investigate the valley properties, the 3D energy band map in the first Brillouin zone is shown in Fig. S5a of the ESI.† By projecting the two energy bands of the VBM and CBM onto the horizontal plane, as shown in Fig. S5b and c of the ESI,† the energy of the VBM at the K and K′ points is not a global but a local minimum, while the energy of the CBM at the K and K′ points is a global maximum. The valley polarization of the Janus 2H-GdIBr monolayer is much larger than those of other reported ferro-valley materials, *e.g.*, GdF_2 (47.6 meV),[43] $VAgP_2Se_6$ (15 meV),[13] and $LaBr_2$ (33 meV).[44] To be a valley material for practical applications, the valley polarization must be large enough. Generally, an estimated valley polarization of 100 meV is required to overcome the thermal noise.[45] In addition, the absence of other bands within the transmission energy window of −2 to 0 eV provides more possibilities for tuning the valley polarization of the Janus 2H-GdIBr monolayer, such as applying magnetic fields and changing the magnetic axis orientation.

To ensure the accuracy of Wannier basis functions, the tightly bound band structure with the spin operator projection $\hat{S}_z$, obtained from the Wannier interpolation,[46] is calculated using MLWFs [Fig. 2c], which is basically in full agreement with the DFT results [Fig. 2b]. The average spin values of the valence and conduction band edge states are found to be 1 and −1, respectively. Thus, their spins remain almost parallel in the upward and downward directions, which usually generates an effective magnetic field that contributes to the energy transfer. Simultaneously, the simulated ARPES spectrum [Fig. 2d] is calculated using the Wannier function and the iterative Green's function method,[47,48] strongly suggesting that the surface is also in a semiconductor state.

Magnetic anisotropy is a critical factor for realizing the long-range FM ordering and the corresponding magnetic anisotropy energy (MAE) can be calculated by considering the transition of magnetic moment from the in-plane [100] to the out-of-plane [001] axis. The Janus 2H-GdIBr monolayer exhibits in-plane magnetic anisotropy (IMA) with a value of 0.42 meV, which is between those of the $2H\text{-}GdI_2$ (0.688 meV) and $2H\text{-}GdBr_2$ (0.109 meV) monolayers.[19] Fig. 3a shows the MAE of the Janus 2H-GdIBr monolayer under various strains. The in-plane strain can be denoted as $\varepsilon = (a - a_0)/a_0 \times 100\%$, where a and a_0 represent the in-plane lattice constants of the strained and unstrained Janus 2H-GdIBr monolayers, respectively. It is obvious that the MAE of the Janus 2H-GdIBr monolayer shows a monotonous increase as the strain increases from −8% to 8%. The atomic layer-resolved MAEs for the Janus 2H-GdIBr monolayer under various strains are shown in Fig. 3b. One can see that the MAE of the Janus 2H-GdIBr monolayer is mainly determined by the Gd and I atoms. The Br and I atoms show the IMA character, while the Gd atom exhibits a transition from the PMA to IMA characters as the strain increases from −8% to 8%. The orbital-resolved MAE can be defined by using the second-order perturbation theory:[49]

$$\mathrm{MAE} = \xi^2 \sum_{\mathrm{o,u}} \frac{\left|\langle\psi_\mathrm{o}|\hat{L}_x|\psi_\mathrm{u}\rangle\right|^2 - \left|\langle\psi_\mathrm{o}|\hat{L}_z|\psi_\mathrm{u}\rangle\right|^2}{E_\mathrm{u} - E_\mathrm{o}} \tag{1}$$

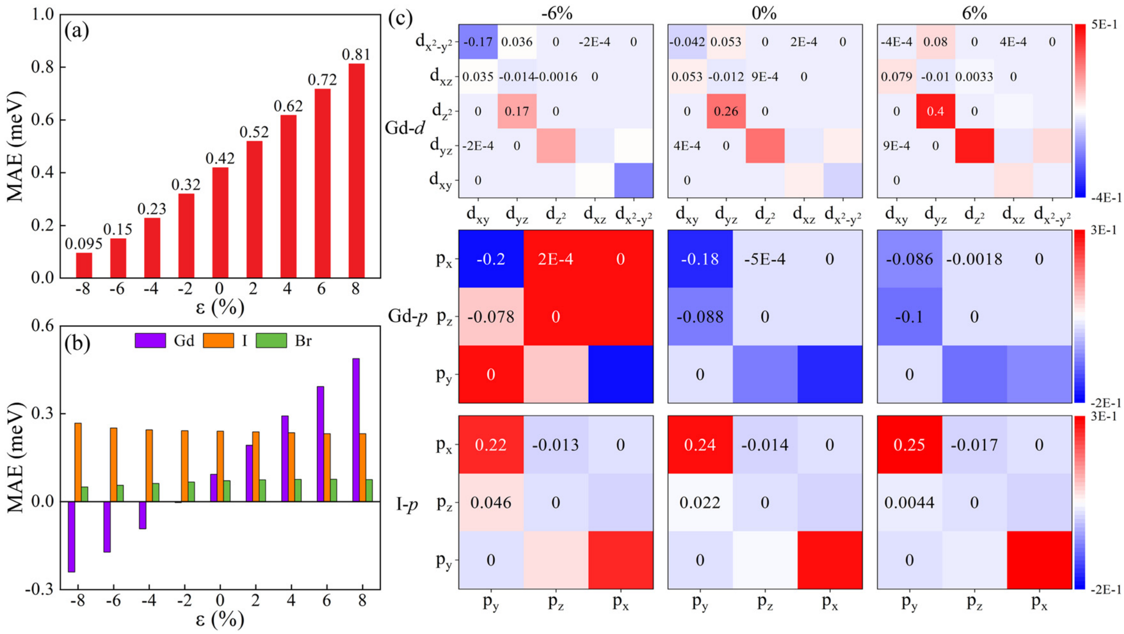

Fig. 3 (a) Total and (b) atomic-layer-resolved MAEs for the Janus 2H-GdIBr monolayer under various strains. (c) Gd-d, Gd-p and I-p orbital resolved MAEs for the Janus 2H-GdIBr monolayer under biaxial strains of −6%, 0% and 6%.

where o/u (E_o/E_u) represent the index (energy) of occupied/non-occupied states and ξ represents the SOC constant. The Gd-d, Gd-p and I(Br)-p orbital resolved MAEs of the Janus 2H-GdIBr monolayer at strains of −6%, 0%, and 6% are shown in Fig. 3c and Fig. S6 of the ESI.† For the unstrained (0%) Janus 2H-GdIBr monolayer, more significant positive MAE (IMA) can be attributed to the matrix element difference (p_x, p_y) of the I atom as well as the matrix element differences (d_{z^2}, d_{yz}) and (p_x, p_y) of the Gd atom. When the strain changes from −6% to 6%, the positive MAE increases, which is mainly caused by the increase of the matrix element difference (d_{z^2}, d_{yz}) of the Gd atom and the decrease of the matrix element differences (p_x, p_y) and ($d_{x^2-y^2}$, d_{xy}) of the Gd atom. It is noted that the matrix element differences (d_{z^2}, d_{yz}) and ($d_{x^2-y^2}$, d_{xy}) of the Gd atom show the opposite signs at a strain of −6%. However, at the strain of 6%, the matrix element difference ($d_{x^2-y^2}$, d_{xy}) of the Gd atom remarkably decreases to negligible while the matrix element difference (d_{z^2}, d_{yz}) of the Gd atom continue to increase and become dominant. Meanwhile, the matrix element differences (p_x, p_y) of I and Br atoms remain stable, and the matrix element differences (p_x, p_y) of the Gd atom monotonically decrease, resulting in the appearance of an IMA character. Thus, the matrix element differences (d_{z^2}, d_{yz}), ($d_{x^2-y^2}$, d_{xy}) and (p_x, p_y) of Gd atoms dominate the MAE behaviors under various strains for the Janus 2H-GdIBr monolayer. The dependence of MAE (MAE = $E^{[001]} - E^{\theta}$) on the polar angle (θ) is also investigated. Fig. 4a shows the angular dependence of MAE along the X (100), Y (010), and Z (001) axes. It is clear that the MAE strongly depends on the direction of magnetization in the xz and yz planes. In contrast, the MAE in the xy plane is isotropic. Thus, a strong dependence of MAE on the angle of magnetization in the out-of-plane is observed, which is similar to the case of VS_2 [50] and $FeCl_2$ [51] monolayers. The strong magnetic anisotropy of the Janus 2H-GdIBr monolayer can be confirmed by the distribution of MAE on the whole space, as shown in Fig. 4b. The MAE is zero in the xy plane and reaches a maximum value of 415 μeV per Gd in the xz (yz) plane, which is comparable to that of the GdI_2 monolayer (553 μeV per Gd).[19]

A high T_c is crucial for the practical application of spintronic devices. To determine the long-range magnetic exchange interactions of the Janus 2H-GdIBr monolayer, the metropolis Monte Carlo algorithm based on the Heisenberg model is used, and the spin Hamiltonian can be described as[52]

$$H = -J\sum_{i,j} S_i S_j - D\sum_i (S_i^{\,z})^2 \tag{2}$$

where S_i and S_j are the spin operators, J represents the isotropic part of exchange interaction, and D represents the single-ion magnetic anisotropy on iron. All parameters in eqn. (2) are determined by the energy mapping in relativistic generalized density functional theory calculations for the 2 × 2 × 1 supercells, including cells with the FM and AFM configurations. The energies of FM and AFM states and the magnetic ion anisotropy parameter D by considering the SOC effect can be calculated as follows:

$$E_{\mathrm{FM}} = E_0 - 6J|S|^2 - D|S|^2 \tag{3}$$

$$E_{\mathrm{AFM}} = E_0 - 2J|S|^2 - D|S|^2 \tag{4}$$

where E_{FM} and E_{AFM} are the total energies of FM and AFM configurations. E_0 is the total energy in the system without mag-

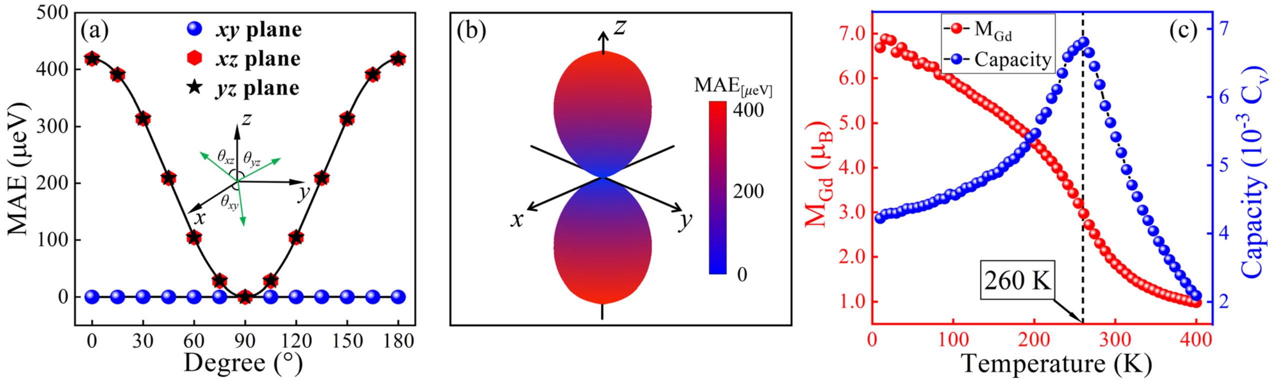

Fig. 4 Dependence of MAE on the polar angle (θ) for the Janus 2H-GdIBr monolayer with the direction of magnetization lying on (a) the *xy*, *xz* and *yz* planes as well as (b) the whole space. (c) Evolution of the average magnetic moment and specific heat for the Janus 2H-GdIBr monolayer.

netic exchange coupling. $|S|$ is the total spin of Gd. The nearest neighbor exchange parameter J is defined as

$$J = \frac{E_{\text{AFM}} - E_{\text{FM}}}{8|S|^2}. \quad (5)$$

The calculated J value for the Janus 2H-GdIBr monolayer is 1.188 meV, suggesting that the magnetic interactions between the nearest neighboring Gd atoms are in the FM state. Fig. 4c shows the temperature dependences of average magnetic moment and specific heat for the Janus 2H-GdIBr monolayer. During the simulation from 0 K to 400 K, T_c can be estimated to be about 260 K. Combined with the calculated high T_c, the Janus 2H-GdIBr monolayer can be expected to be an ideal ferro-valley material for applications in valley electronics. A real-space renormalization group analysis is used to avoid errors originating from finite size by roughly dividing the original 32 × 32 lattices into a 16 × 16 lattice, with every four adjacent spins forming quasiparticles, each with a representative spin. The Heisenberg Hamiltonian of the renormalized model is assumed to be the same as before, and the internal energy of this system is recalculated. More details are shown in Fig. S7 of the ESI.†

The highly localized 4f electrons on Gd atoms are negligible in the magnetic exchange coupling. However, more extended 5d electrons on Gd atoms occupy the spin-up channel near E_F, making the Janus 2H-GdIBr monolayer a magnetic semiconductor. According to the Kramer mechanism, the partially occupied states can result in the direct exchange interaction between the nearest Gd–Gd atoms, and the bond angle of Gd–I (Br)–Gd is close to 90°, satisfying the Goodenough–Kanamori–Anderson (GKA) rules.[53–55] To investigate the origin of FM coupling in the Janus 2H-GdIBr monolayer, inter-atomic interactions need to be considered. Generally, the direct (super) exchange depends on the Gd–Gd (Gd–I(Br)–Gd) bond, which leads to the FM (AFM) states. Generally, the Gd–Gd (I/Br) distance and the Gd–I(Br)–Gd bond angle mainly contribute to this magnetic exchange mechanism. Thus, the synergistic effects of direct and super exchange interactions determine the magnetic ground state of the Janus 2H-GdIBr monolayer.

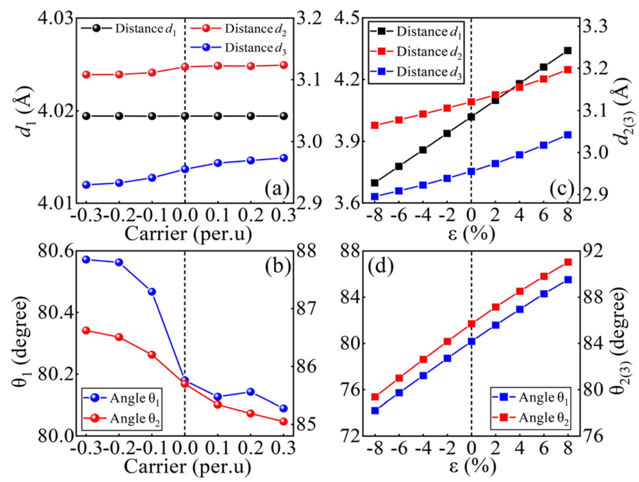

Fig. 5 Charge carrier doping dependence of the inter-atomic (a) distance and (b) angle for the Janus 2H-GdIBr monolayer. Biaxial strain dependence of the inter-atomic (c) distance and (d) angle for the Janus 2H-GdIBr monolayer.

The high concentration of hole carrier doping usually induces a transition from the AFM to FM states, proving the effectiveness of modulating the magnetic behavior by charge carrier doping. As shown in Fig. 5a and b, the nearest Gd–Gd distance d_1 always remains constant, and the Gd–I distance d_2 and Gd–Br distance d_3 show a monotonic and slightly increasing (decreasing) trend under electron (hole) doping. Meanwhile, the Gd–I–Gd bond angle θ_1/Gd–Br–Gd bond angle θ_2 increases (decreases) with increasing hole (electron) concentration, which significantly strengthens (weakens) the super-exchange interaction. During the experiment, it is common to have various strains caused by the lattice mismatch of the substrate. Fig. 5c and d show the dependence of the Gd–Gd(I/Br) distance d and the Gd–I(Br)–Gd bond angle θ on strains. As the strain increases from −8% to 8%, the nearest Gd–Gd distance d_1 (from 3.70 Å to 4.34 Å) and Gd–I–Gd bond angle θ_1 (from 74.2° to 85.5°)/Gd–Br–Gd bond angle θ_2 (from 79.4° to 91.0°) exhibit monotonically increasing behaviors. According to the

GKA rule, the increased (decreased) d_1 weakens (strengthens) the direct exchange interaction. Meanwhile, the increased (decreased) angle θ_1/θ_2 enhances (weakens) the super-exchange interaction. Clearly, the direct exchange and super-exchange interactions compete with each other. Combining with the dependence of the exchange parameter J on strain [Fig. S8a of the ESI†], the increased or decreased d_1 makes the direct exchange interaction more critical in determining the FM state. Generally, the tensile (compressive) strain weakens (enhances) the FM coupling and plays a vital role in the direct and superexchange mechanisms. Meanwhile, in the range of hole doping, the exchange parameter J changes slightly, and the T_c near room-temperature is well preserved for the Janus 2H-GdIBr monolayer. In contrast, the exchange parameter J and corresponding T_c change significantly with the increase in electron doping [Fig. S8b–d of the ESI†]. The tensile (compressive) strain tends to monotonically decrease (increase) the exchange parameter J and corresponding T_c. Interestingly, the maximum T_c reaches 281 K at a tensile strain of 8%, which is close to the room temperature [Fig. S8a, c and d of the ESI†].

When the space-inversion symmetry is broken in hexagonal systems, the charge carriers at the K and K′ valleys can generate non-zero Berry phases along the z-direction, accompanied by a non-zero Berry curvature. When the time-reversal symmetry is also broken, the characteristic of valley contrast appears. To investigate this property, the Berry curvature of the Janus 2H-GdIBr monolayer is calculated according to the Kubo formula:[56]

$$\Omega(k) = -\sum_n \sum_{n\neq m} f(n) \frac{2\mathrm{Im}\langle\varphi_{nk}|v_x|\varphi_{mk}\rangle\langle\varphi_{mk}|v_y|\varphi_{nk}\rangle}{(E_n - E_m)^2} \tag{6}$$

where υ_x and υ_y are the velocity operators of the Dirac electrons along the x and y directions, respectively. $f(n)$ is the Fermi–Dirac distribution function for the nth band, and $|\varphi_{nk}$ is the calculated Bloch wave function with the energy eigenvalue $E_n(E_m)$. The maximally localized Wannier function is used to calculate the Berry curvature. Fig. 6a and b show the Berry curvature of the Janus 2H-GdIBr monolayer along the high symmetry line and over the 2D Brillouin zone. It is clear that the K and K′ valleys have opposite signs of the Berry curvature and slightly different absolute values, revealing that the valley contrast phenomenon plays an important role in characterizing the Bloch electron chirality in the k-space.[57] Away from the K and K′ valleys, the Berry curvature decays rapidly and disappears at the M point. As described in eqn. (7),[58] the Berry curvature drives a peculiar transverse velocity in the presence of an in-plane electric field E:

$$\nu = -\frac{e}{\hbar} E \times \Omega(k). \tag{7}$$

This is an intrinsic contribution of the anomalous Hall effect. The integral of the Berry curvature in the Brillouin zone contributes to the anomalous Hall conductivity (AHC), which can be defined as[59]

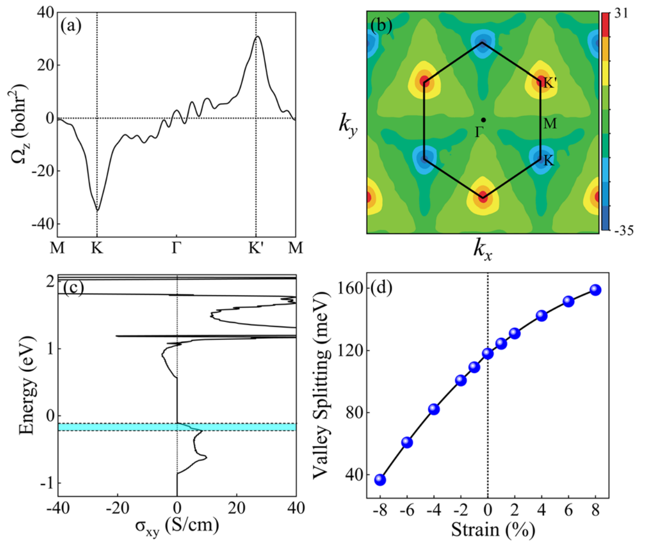

Fig. 6 Calculated Berry curvature of the Janus 2H-GdIBr monolayer (a) along high symmetry lines and (b) over the 2D Brillouin zone. (c) Calculated anomalous Hall conductivity of the Janus 2H-GdIBr monolayer; the two dashed lines denote the two valley extrema. (d) Variation of valley polarization for the Janus 2H-GdIBr monolayer as a function of strain from −8% to 8%.

$$\sigma_{xy} = -\frac{e^2}{\hbar}\int_{\mathrm{BZ}} \frac{d^2k}{(2\pi)^2}\Omega(k). \tag{8}$$

Due to the opposite sign and unequal absolute value of the Berry curvature, the charge carriers in the K and K′ valleys have opposite transverse velocities. Thus, anomalous Hall conductivity with a net value is generated. As shown in Fig. 6c, when the E_F is located between the valence band edges of the K and K′ valleys, a fully valley-polarized Hall conductivity is generated with a calculated maximum AHC of 9.5 S cm^{-1}. In addition, the E_F can be effectively tuned above (below) the valley splitting gap by a low concentration of carrier doping [Fig. S9 of the ESI†]. On the other hand, if a finite valley polarization can be generated, for example, by irradiating with circularly polarized light [see details in Fig. S10 of the ESI†], the Hall currents will also appear. As shown in Fig. 6d, the valley polarization increases monotonically with the increase (decrease) of tensile (compressive) strains and a giant valley polarization of about 158 meV is obtained at a tensile strain of 8%. But the valley polarization remains significant (about 37 meV) under a considerable compressive strain of −8%, implying that the valley polarization is robust against the in-plane strain. Therefore, the Janus 2H-GdIBr monolayer can be considered as an ideal ferro-valley material providing a potential platform for valley electronic applications.

4. Conclusions

In summary, by first principles calculations, we predict that the Janus 2H-GdIBr monolayer exhibits a low exfoliation energy of 0.24 J m^{-2} and possesses tunable semiconductor characteristics with a sizeable magnetic moment, a high T_c of 260 K, in-plane magnetic anisotropy, and excellent thermal and dynamic stability. The MAE and valley polarization features are robust against the in-plane biaxial strains. The total MAE can increase monotonically from nearly 0.1 to 0.81 meV per f.u. under a strain from −8% to 8%. Meanwhile, the easy axis of the Gd atom shows a transition from the [001] to [100] direction. Based on the second-order perturbation theory, it is found that the competition between the contributions of the Gd-d orbitals, Gd-p orbitals, and p orbitals of halogen atoms induce the IMA character. When a tensile strain of 8% is applied, the valley polarization of the Janus 2H-GdIBr monolayer can reach a maximum value of 158 meV. Furthermore, the corresponding T_c is tuned by strains from 233 K (8%) to 281 K (−8%) and charge carrier doping from 140 K (0.3 e per f.u.) to 245 K (−0.3 e per f.u.). In addition, the origin of FM coupling in the Janus 2H-GdIBr monolayer can be attributed to the direct-exchange (Gd–Gd) and super-exchange (Gd–I(Br)–Gd) interactions. Due to the space-inversion and time-reversal symmetry breaking, an appropriate external electric field E can achieve the anomalous valley Hall effect in the Janus 2H-GdIBr monolayer. Overall, our findings have created expectation that the Janus-2H-GdIBr monolayer is a candidate material for the next-generation of practical spintronic/valleytronic devices.

Author contributions

Cunquan Li: data curation, investigation, and writing – original draft. Yukai An: supervision and writing – reviewing and editing.

Conflicts of interest

There are no conflicts of interest to declare.

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